

LAPORAN PRAKTIKUM
BASIS DATA

MODUL 8 STUDY CASE



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1. CREATE TABLE

```
CREATE TABLE department (  
    department_id NUMBER(4) PRIMARY KEY,  
    department_name VARCHAR2(50),  
    manager_id NUMBER(6),  
    location_id NUMBER(4)  
);
```

```
CREATE TABLE employee (  
    employee_id NUMBER(6) PRIMARY KEY,  
    last_name VARCHAR2(50) NOT NULL,  
    email VARCHAR2(50),  
    salary NUMBER(8,2),  
    commission_pct NUMBER(4,2),  
    hire_date DATE NOT NULL,  
    department_id NUMBER(4),  
  
    CONSTRAINT fk_employee_department  
    FOREIGN KEY (department_id)  
    REFERENCES department(department_id)  
);
```

```
CREATE TABLE supplier (  
    supplier_id NUMBER(4) PRIMARY KEY,  
    supplier_name VARCHAR2(50),  
    phone VARCHAR2(20)  
);
```

```
CREATE TABLE inventory (  
    item_id NUMBER(4) PRIMARY KEY,  
    item_name VARCHAR2(50),  
    stock NUMBER(5),  
    price NUMBER(10,2),  
    supplier_id NUMBER(4),  
  
    CONSTRAINT fk_inventory_supplier  
    FOREIGN KEY (supplier_id)  
    REFERENCES supplier(supplier_id)  
);
```

Script Output x

Task completed in 0.054 seconds

Table DEPARTMENT created.

Table EMPLOYEE created.

Table SUPPLIER created.

Table INVENTORY created.

2. ANALISIS COMPUTE STATISTIC

```
ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE supplier COMPUTE STATISTICS;
ANALYZE TABLE inventory COMPUTE STATISTICS;
```

Script Output x

Task completed in 0.167 seconds

Table DEPARTMENT analyzed.

Table EMPLOYEE analyzed.

Table SUPPLIER analyzed.

Table INVENTORY analyzed.

3. INSERT DATA

```
ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE supplier COMPUTE STATISTICS;
ANALYZE TABLE inventory COMPUTE STATISTICS;

INSERT INTO department VALUES (10, 'IT', 101, 1000);
INSERT INTO department VALUES (20, 'HR', 102, 1001);
INSERT INTO department VALUES (30, 'Finance', 103, 1002);
INSERT INTO department VALUES (40, 'Marketing', 104, 1003);
INSERT INTO department VALUES (50, 'Sales', 105, 1004);
INSERT INTO department VALUES (60, 'Production', 106, 1005);
INSERT INTO department VALUES (70, 'Support', 107, 1006);
INSERT INTO department VALUES (80, 'Administration', 108, 1007);
INSERT INTO department VALUES (90, 'Logistic', 109, 1008);
INSERT INTO department VALUES (100, 'Research', 110, 1009);

INSERT INTO employee VALUES (1001, 'Andi', 'andi@mail.com', 5000, 0.10, DATE '2023-01-01', 10);
INSERT INTO employee VALUES (1002, 'Budi', 'budi@mail.com', 5500, 0.05, DATE '2023-01-02', 20);
INSERT INTO employee VALUES (1003, 'Citra', 'citra@mail.com', 6000, 0.07, DATE '2023-01-03', 30);
INSERT INTO employee VALUES (1004, 'Dina', 'dina@mail.com', 6500, 0.06, DATE '2023-01-04', 40);
INSERT INTO employee VALUES (1005, 'Eko', 'eko@mail.com', 7000, 0.08, DATE '2023-01-05', 50);
INSERT INTO employee VALUES (1006, 'Farah', 'farah@mail.com', 7500, 0.04, DATE '2023-01-06', 60);
INSERT INTO employee VALUES (1007, 'Gilang', 'gilang@mail.com', 8000, 0.09, DATE '2023-01-07', 70);
INSERT INTO employee VALUES (1008, 'Hani', 'hani@mail.com', 8500, 0.05, DATE '2023-01-08', 80);
INSERT INTO employee VALUES (1009, 'Indra', 'indra@mail.com', 9000, 0.10, DATE '2023-01-09', 90);
INSERT INTO employee VALUES (1010, 'Joko', 'joko@mail.com', 9500, 0.03, DATE '2023-01-10', 100);
```

```

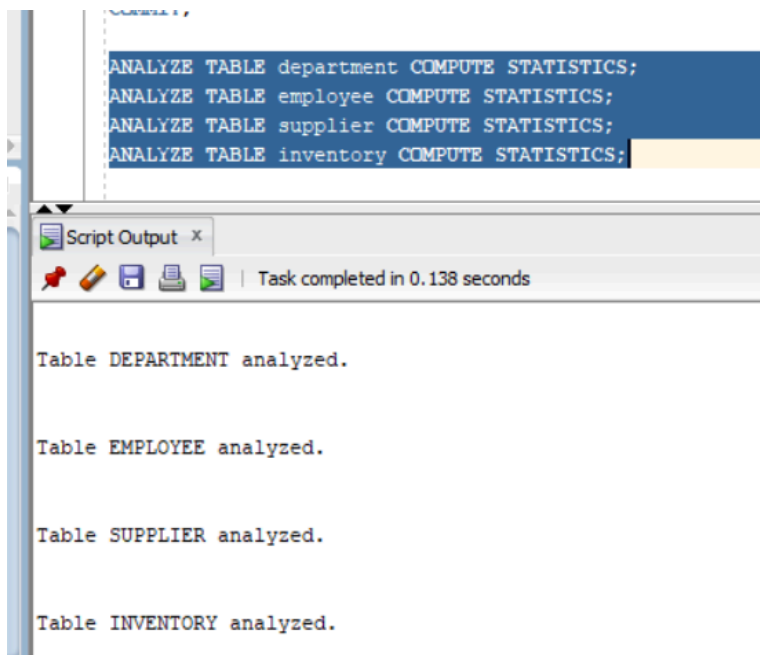
INSERT INTO supplier VALUES (1,'PT Maju Jaya','0811111111');
INSERT INTO supplier VALUES (2,'PT Sentosa','0822222222');
INSERT INTO supplier VALUES (3,'PT Nusantara','0833333333');
INSERT INTO supplier VALUES (4,'PT Berkah','0844444444');
INSERT INTO supplier VALUES (5,'PT Sejahtera','0855555555');
INSERT INTO supplier VALUES (6,'PT Global','0866666666');
INSERT INTO supplier VALUES (7,'PT Prima','0877777777');
INSERT INTO supplier VALUES (8,'PT Mandiri','0888888888');
INSERT INTO supplier VALUES (9,'PT Abadi','0899999999');
INSERT INTO supplier VALUES (10,'PT Makmur','0800000000');

INSERT INTO inventory VALUES (101,'Laptop',15,12000000,1);
INSERT INTO inventory VALUES (102,'Printer',10,2500000,2);
INSERT INTO inventory VALUES (103,'Mouse',50,150000,3);
INSERT INTO inventory VALUES (104,'Keyboard',40,300000,4);
INSERT INTO inventory VALUES (105,'Monitor',20,2000000,5);
INSERT INTO inventory VALUES (106,'Scanner',8,1800000,6);
INSERT INTO inventory VALUES (107,'Router',12,800000,7);
INSERT INTO inventory VALUES (108,'Switch',10,1500000,8);
INSERT INTO inventory VALUES (109,'Harddisk',30,900000,9);
INSERT INTO inventory VALUES (110,'SSD',25,1200000,10);

```

COMMIT;

4. ANALISIS



5. View empvu80

The screenshot shows the SQL Developer interface. The top pane contains the SQL script to create a view named empvu80. The script selects employee_id (aliased as id_number), last_name (aliased as name), salary, and department_id from the employee table, filtered by department_id = 80. The bottom pane shows the script output, indicating that the tables EMPLOYEE, SUPPLIER, and INVENTORY were analyzed, and the view EMPVU80 was successfully created. Below the output, the same SQL script is shown again, but with an additional query: SELECT * FROM empvu80;.

```
CREATE VIEW empvu80 AS
SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM employee
WHERE department_id = 80;
```

Script Output x
Task completed in 0.072 seconds

Table EMPLOYEE analyzed.
Table SUPPLIER analyzed.
Table INVENTORY analyzed.
View EMPVU80 created.

```
CREATE VIEW empvu80 AS
SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM employee
WHERE department_id = 80;

SELECT * FROM empvu80;
```

Script Output x
Task completed in 0.055 seconds

Table EMPLOYEE analyzed.
Table SUPPLIER analyzed.
Table INVENTORY analyzed.
View EMPVU80 created.

EMPLOYEE_ID	NAME	SALARY	DEPARTMENT_ID
1008	Hani	8500	80

6. STRUKTUR EMPUV

```
SELECT * FROM empvu80;
DESC empvu80;
```

Script Output x
Task completed in 0.428 seconds

VIEW EMPVU80 created.

ID_NUMBER	NAME
1008	Hani

Name	Null?	Type
ID_NUMBER	NOT NULL	NUMBER(6)
NAME	NOT NULL	VARCHAR2(50)
SALARY		NUMBER(8,2)
DEPARTMENT_ID		NUMBER(4)

7. NON UNIQUE INDEX

```
CREATE INDEX idx_emp_department
ON employee(department_id);

SELECT index_name,
       table_name
FROM user_indexes
WHERE table_name='EMPLOYEE';
```

Script Output x
Task completed in 0.083 seconds

NAME	NOT NULL	VARCHAR2(50)
SALARY		NUMBER(8,2)
DEPARTMENT_ID		NUMBER(4)

Index IDX_EMP_DEPARTMENT created.

INDEX_NAME	TABLE_NAME
SYS_C008433	EMPLOYEE
IDX_EMP_DEPARTMENT	EMPLOYEE